PROJECT DESIGN PHASE-I

Problem – Solution Fit Template

Date	28 October 2026	Maximum Marks	2 Marks
Team ID	EN25CS3011166	Project Name	Health360: Cardiovascular Risk Analytics

Problem – Solution Fit Canvas: The Problem-Solution Fit helps identify behavioral patterns and verify that the proposed Business Intelligence solution solves key user pain points effectively.

- Solve complex cardiovascular risk analysis problems aligned with user workflows.
- Succeed faster by tapping into existing diagnostic data channels and clinical routines.
- Increase touch-points with preventive healthcare insights and build user trust.

DEFINE CS, FIT INTO CC

1. CUSTOMER SEGMENT(S) — CS

Who is your customer?

- **Primary:** Individuals aged 19–79 concerned about hypertension, heart disease, or stroke risk.
- **Secondary:** Cardiologists, clinicians, and health coaches monitoring patient cohorts.
- **Tertiary:** Healthcare administrators managing preventive care analytics.

6. CUSTOMER CONSTRAINTS — CC

What constraints limit their choices?

- Limited time to interpret raw lab test values.
- Lack of medical knowledge to calculate composite risk scores.
- Privacy concerns regarding sensitive health metrics.
- Device variability (mobile vs desktop access).

5. AVAILABLE SOLUTIONS — AS

What existing solutions exist & pros/cons?

- **Paper Blood Pressure Logs:** Easy, but lacks predictive trends or correlations.
- **Generic Fitness Apps:** Tracks steps/calories, but lacks clinical stroke & cardiac risk mapping.
- **Hospital EHR Systems:** Robust data, but complex interface and restricted patient access.

EXPLORE AS, DIFFERENTIATE

FOCUS ON J&P, TAP INTO BE, UNDERSTAND RC

2. JOBS-TO-BE-DONE / PROBLEMS — J&P

Which jobs/problems do you address?

- Assess combined cardiovascular risk from cholesterol, BP, glucose, and BMI.
- Identify how lifestyle factors (smoking, alcohol, fatigue) impact health.
- Detect early warning signs of chest pain and dizziness.

9. PROBLEM ROOT CAUSE — RC

What is the real reason this exists?

- Health data is fragmented across diagnostic labs, fitness wearables, and EHRs.
- Cardiovascular risk is multi-factorial; isolated metrics miss compound risk factors.
- Late diagnosis due to asymptomatic early stages of hypertension and high cholesterol.

7. BEHAVIOUR — BE

What does customer do to address it?

- Periodically checks BP at local pharmacies or home monitors.
- Searches online symptoms when experiencing fatigue or chest tightness.
- Consults doctors only when symptoms become acute or severe.

FOCUS ON J&P, TAP INTO BE, UNDERSTAND RC

IDENTIFY STRONG TR & EM

3. TRIGGERS — TR

What triggers customers to act?

- Routine annual health checkup showing elevated readings.
- Experiencing fatigue, dizziness, or chest discomfort.
- Family history of heart disease or stroke.

4. EMOTIONS: BEFORE / AFTER — EM

How do customers feel?

- **Before:** Anxious, confused by lab numbers, uncertain about stroke risk.
- **After:** Informed, confident, in control with clear visual risk heatmaps.

10. YOUR SOLUTION — SL

How does your solution fit customer needs?

Health360 Cardiovascular BI Platform:

- **Interactive Dashboard:** Consolidates 8 critical risk charts (Age Trends, BMI vs Heart Disease, BP Risk, Glucose, Alcohol/Smoking).
- **Patient Risk Heatmap:** Instant visual categorization of high-risk cohorts.
- **Personalized Metrics:** Clear percent risk breakdown for exercise level, fatigue, and chest pain indicators.

8. CHANNELS OF BEHAVIOUR — CH

Where do they take action?

8.1 ONLINE

- Web-based Tableau dashboards & Streamlit portals.
- Mobile health applications & email alerts.

8.2 OFFLINE

- Clinician consultations & printed risk summary reports.
- Pharmacy health kiosks.

EXTRACT ONLINE & OFFLINE CH OF BE